

Mohamed Zeina

+90 552 511 1730 | mohamed.zeina.cv@gmail.com | [linkedin.com/in/mohamedzeina](https://www.linkedin.com/in/mohamedzeina) | github.com/mohamedzeina | mohamedzeina.github.io

EXPERIENCE

Embedded Systems & Software Engineer (Contract)

August 2025 – Feb. 2026

Energco, ZAIM Teknopark

Istanbul, TR

- Built a multi-drone system coordinating 5 UAVs with real-time 30 FPS video streaming, live telemetry via WebSockets, and fault-tolerant reconnection and failsafe handling
- Developed a Ground Control Station using React and Python FastAPI with real-time map interface, manual flight controls, and remote GPS navigation
- Calculated target GPS coordinates from mother drone's gimbal angles, GPS, and rangefinder data to autonomously dispatch mini drones
- Implemented YOLOv8 object detection with PID-controlled flight for real-time subject tracking on mini drone video streams
- Configured bridged wireless network between GCS and drones, and set up embedded Linux on Raspberry Pi for video streaming and MAVLink communication

Teaching Assistant for Introduction to Computing

Sep. 2021 – Jan. 2024

Sabanci University

Istanbul, TR

- Tutored over 50 students weekly in C++ programming during recitations and office hours
- Designed and evaluated 30+ assignments and exams for Introduction to Computing (C++), directly aligning assessment criteria with course learning objectives

Full Stack Software Engineer Intern

July 2019 – Sep. 2019

Interact Technologies

Istanbul, TR

- Implemented reliable TCP/IP communication protocol, achieving 99.9% data transmission success over network tests
- Reduced GUI load time by 40% through code optimization
- Designed and implemented a NoSQL document database schema to store and retrieve structured therapy session records, supporting 500+ patient records and enabling downstream reporting
- Designed 8 3D animations that decreased therapy learning time by 25%

EDUCATION

Sabanci University

Istanbul, TR

Master of Science in Computer Science and Engineering (with thesis) (M.Sc.CSE)

Sep. 2021 – Jan. 2024

- GPA: 3.91/4.0
- Full Scholarship
- Thesis: Built a framework combining GANs, SMOTE, and differential privacy to preserve household privacy in smart meter data while maintaining machine learning model utility

Sabanci University

Istanbul, TR

Bachelor of Computer Science and Engineering (B.Sc.CSE)

Sep. 2016 – June 2020

- GPA: 3.73/4.0
- Half Tuition Scholarship
- Degree of High Honor (2016-2017, 2018-2020)

PROJECTS

Node Social | *Node.js, Express.js, MongoDB, GraphQL, JWT, React*

- Built and deployed scalable GraphQL API handling 1,000+ simulated requests/sec with JWT-based access control
- Implemented time-limited token authentication and real-time access control for post management
- Developed image upload functionality with database integration using Multer

Multi-Docker Fibonacci Calculator | *Docker, React, Node.js, Redis, PostgreSQL, AWS*

- Developed a full-stack microservices app to calculate Fibonacci numbers using React (client), Express (API), Redis, and PostgreSQL
- Containerized all components with Docker, orchestrated deployments via Kubernetes, and routed traffic through NGINX
- Implemented CI/CD pipeline with Travis CI to test, build, push Docker images to Docker Hub, and auto-deploy to AWS Elastic Beanstalk
- Used AWS RDS and ElastiCache for production-grade PostgreSQL and Redis; integrated Scaffold for live-reload development

Online Shop | *Node.js, Express.js, MongoDB, MVC, EJS, Stripe*

- Developed a full-stack e-commerce platform with Node.js/Express.js backend and EJS frontend
- Enhanced security with CSRF protection and integrated Stripe payment gateway
- Utilized MongoDB with Mongoose for data management and implemented secure authentication with bcryptjs

JSNote | *React, TypeScript, ESBuild, Redux, Express, Immer, Lerna, CLI*

- Published `jsnote-zeina` to NPM — used by developers as a standalone in-browser code editor
- Created in-browser coding environment with live transpiling/bundling and sandboxed execution
- Managed application state with Redux/Immer and implemented API server with CLI integration

Discuss App | *Next.js, SQLite, Next-Auth, Next-UI, Zod*

- Built a full-stack discussion platform using Next.js with server-side rendering and SQLite as a lightweight embedded DB
- Implemented GitHub OAuth authentication using Next-Auth for secure user login and session management
- Designed responsive, accessible UI using Next-UI components with loading skeletons and modals for improved UX
- Validated user input with Zod and optimized API routes with memoization, reducing query response time by 50%

SKILLS

Programming Languages: C++, C#, Python, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Redux, Next.js, Node.js, Express, Django, Deno, Mongoose, GraphQL, EJS, JWT, OAuth2

Web Technologies: HTML, CSS, Tailwind CSS, RESTful APIs, WebSockets

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

DevOps & Infrastructure: Docker, Nginx, AWS (EC2, RDS, S3, IAM), Google Cloud (GKE)

Developer Tools: Git, GitHub, CI/CD (GitHub Actions, Travis CI)

Spoken Languages: Arabic (Native), English (Fluent)

Embedded & IoT: Raspberry Pi, MAVLink, UDP Streaming, Wireless Networking

Computer Vision: OpenCV, YOLOv8, Object Tracking